Production Deployment Guide (Windows Server)

This guide provides step-by-step instructions to deploy the Insurance Renewal Application on a fresh Windows Server using Docker Desktop.

1. Prerequisites

1. **Remote Access**: Log in to your Windows Server (via RDP).
2. **Docker Desktop**:
   - Download and install Docker Desktop for Windows.
   - Restart the server if required by the installer.
   - Start Docker Desktop and ensure it is running (look for the whale icon in the system tray).
3. **Project Files**:
   - Copy the entire project folder (`insurance-renewal-app`) from your local machine to the server (e.g., to `C:\insurance-renewal-app`).

2. Start the Application

1. Open **PowerShell** or **Command Prompt** as Administrator.
2. Navigate to the project directory:

```
cd C:\insurance-renewal-app
```

(Replace `C:\insurance-renewal-app` with your actual path)

3. Build and start the services:

```
docker-compose up -d --build
```

4. Wait for the application to start. You can check the status with:

```
docker ps
```

Ensure `insurance-backend`, `insurance-frontend`, and `insurance-mysql` are all listed and "Up".

3. Load Data (First Time Only)

Since the database is fresh, you need to load the customer and policy data manually. We will copy the data file into the container and import it to ensure reliability.

1. **Copy the data file to the database container**:
   ```
   docker cp backend/src/main/resources/data.sql insurance-mysql:/tmp/data.sql
   ```
2. **Run the import command**:
   - *Note: This process may take a few minutes for ~19,000 records.*

   ```
   docker exec -i insurance-mysql mysql -uroot -proot insurance_renewal -e "source /tmp/data.sql"
   ```

3. **Verify the import**:
   ```
   docker exec -i insurance-mysql mysql -uroot -proot insurance_renewal -e "SELECT COUNT(*) FROM policies;"
   ```
   You should see a count of approximately 19,000+.

4. Access the Application

1. Open a web browser (Chrome/Edge) on the server.
2. Go to: `http://localhost`
3. **Login**:
   - **Username**: `ShareIndia`
   - **Password**: `ShareIndia@123`

5. Troubleshooting

- **Login Failed / 502 Bad Gateway**:
  - The backend might still be starting. Wait 30 seconds and refresh.
  - Check logs: `docker logs --tail 50 insurance-backend`
- **Database connection error**:
  - Ensure the `insurance-mysql` container is "Healthy".
  - Restart everything: `docker-compose restart`